

Arjuna JEE 2025

Maths

DPP: 6

Trigonometric Equations

Q1 Find the solution set of inequality $\sin x > \frac{1}{2}$

where $0 \leq x \leq 2\pi$

- (A) $(0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, 2\pi)$
 (B) $[\frac{\pi}{6}, \frac{5\pi}{6}]$
 (C) $(\frac{\pi}{6}, \frac{5\pi}{6}) \cup (\pi, \frac{7\pi}{6})$
 (D) $(\frac{\pi}{6}, \frac{5\pi}{6})$

Q2 Solve the following inequality $\cos x \leq -\frac{1}{2}$

where $x \in [0, 2\pi]$

- (A) $(0, \frac{2\pi}{3}) \cup (\frac{4\pi}{3}, \pi)$
 (B) $(0, \frac{\pi}{3}) \cup [\frac{2\pi}{3}, \frac{5\pi}{3}]$
 (C) $[\frac{2\pi}{3}, \frac{4\pi}{3}] \cup (0, \frac{\pi}{3})$
 (D) $[\frac{2\pi}{3}, \frac{4\pi}{3}]$

Q3 Solution set of the inequality $\sin x > \frac{1}{\sqrt{2}}$ belongs

- (A) $(2n\pi + \frac{\pi}{4}, 2n\pi + \frac{3\pi}{4})$ where $n \in \mathbb{Z}$
 (B) $(2n\pi - \frac{3\pi}{4}, 2n\pi - \frac{\pi}{4})$ where $n \in \mathbb{Z}$
 (C) $(2n\pi + \frac{\pi}{4}, 2n\pi + \frac{\pi}{2})$ where $n \in \mathbb{Z}$
 (D) None of these

Q4 If $2\cos x < \sqrt{3}$ and $x \in [-\pi, \pi]$, then the solution set for x is

- (A) $[-\pi, \frac{-\pi}{6}) \cup (\frac{\pi}{6}, \pi]$
 (B) $(\frac{-\pi}{6}, \frac{\pi}{6})$
 (C) $[-\pi, \frac{-\pi}{6}] \cup [\frac{\pi}{6}, \pi]$
 (D) NOT

Q5 Solve the following inequality $\tan x \geq -\sqrt{3}$ where $x \in [0, \pi]$

- (A) $(0, \frac{2\pi}{3}]$
 (B) $(\frac{\pi}{3}, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \frac{2\pi}{3})$
 (C) $[0, \frac{\pi}{2}) \cup [\frac{2\pi}{3}, \frac{3\pi}{2})$

(D) $[0, \frac{\pi}{2}) \cup [\frac{2\pi}{3}, \pi]$

Q6 If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2 \theta - 5\sin \theta + 2 > 0$ is

- (A) $(0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, 2\pi)$
 (B) $(\frac{\pi}{8}, \frac{5\pi}{6})$
 (C) $(0, \frac{\pi}{8}) \cup (\frac{\pi}{6}, \frac{5\pi}{6})$
 (D) $[0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, 2\pi]$

Q7 The number of solutions of $\sum_{r=1}^6 \cos(rx) = 6$ in $(0, 2\pi]$ is

- (A) 0
 (B) 1
 (C) 6
 (D) 12

Q8 The number of solutions of x in the interval $[-\pi, \pi]$ of the equation $(1 + \cot 267^\circ)(1 + \tan 222^\circ) = \sec^2 x + \cos^2 x$ is

- (A) 1
 (B) 2
 (C) 3
 (D) 4

Q9 If $\sin x = \lambda$ has exactly one solution in $[0, 9\pi/4]$ then the number of values of λ , is

- (A) 0
 (B) 1
 (C) 2
 (D) 3

Q10 The number of values of $\theta \in [0, 2\pi]$ satisfying $r \sin \theta = \sqrt{3}$ and $r + 4 \sin \theta = 2(\sqrt{3} + 1)$ is

- (A) 4
 (B) 5
 (C) 6
 (D) 7

Q11 The general solution of $8 \tan^2 \frac{x}{2} = 1 + \sec x$ is

- (A) $x = 2n\pi \pm \cos^{-1}(\frac{-1}{3})$
 (B) $x = 2n\pi \pm \frac{\pi}{6}$



(C) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(D) None of these

Q12 The general solution of the equation $\frac{\tan x}{\tan 2x} + \frac{\tan 2x}{\tan x} + 2 = 0$ is

(A) $n\pi$

(B) $\frac{n\pi}{3}$

(C) $(2n - 1)\frac{\pi}{3}, n \in \mathbb{I}$

(D) $(3n \pm 1)\frac{\pi}{3}, n \in \mathbb{I}$

Q13 The general solution of the equation $\tan x + \tan 2x + \tan x \cdot \tan 2x = 1$ is

(A) $x = n\pi + \frac{\pi}{12}, n\pi + \frac{5\pi}{12}, n \in \mathbb{I}$

(B) $x = \frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{I}$

(C) $x = n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{12}, n \in \mathbb{I}$

(D) $x = n\pi - \frac{\pi}{4}, n\pi + \frac{5\pi}{12}, n \in \mathbb{I}$

Q14 If $\sqrt{3} \tan 2\theta + \sqrt{3} \tan 3\theta + \tan 2\theta \tan 3\theta = 1$, then the general value of θ is

(A) $n\pi + \frac{\pi}{5}$

(B) $(n + \frac{1}{6})\frac{\pi}{5}$

(C) $(2n \pm \frac{1}{6})\frac{\pi}{5}$

(D) $(n + \frac{1}{3})\frac{\pi}{5}$

Q15 The set of value of x for which the expression $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1$ is

(A) ϕ

(B) $\pi/4$

(C) $(n\pi + \frac{\pi}{4}; n = 1, 2, 3 \dots)$

(D) $(2n\pi + \frac{\pi}{4}; n = 1, 2, 3 \dots)$

Q16 If $\sec \theta + 1 = (2 + \sqrt{3}) \tan \theta$, then

(A) $\theta = n\pi \pm \frac{\pi}{2}$

(B) $\theta = 2n\pi \pm \frac{\pi}{6}$

(C) $\theta = n\pi \pm \frac{\pi}{4}$

(D) None of these

Q17 Number of solutions of the equation $\cot(\theta) + \cot(\theta + \frac{\pi}{3}) + \cot(\theta - \frac{\pi}{3}) + \cot(3\theta) = 0$ where $\theta \in (0, \frac{\pi}{2})$

(A) Infinite

(B) 0

(C) 1

(D) None of these

Q18 If $\sin(\frac{\pi}{4} \cot \theta) = \cos(\frac{\pi}{4} \tan \theta)$, then $\theta =$

(A) $n\pi + \frac{\pi}{4}$

(B) $2n\pi \pm \frac{\pi}{4}$

(C) $n\pi - \frac{\pi}{4}$

(D) $2n\pi \pm \frac{\pi}{6}$



Answer Key

Q1 (D)
Q2 (D)
Q3 (A)
Q4 (A)
Q5 (D)
Q6 (A)
Q7 (B)
Q8 (C)
Q9 (B)

Q10 (A)
Q11 (C)
Q12 (D)
Q13 (B)
Q14 (B)
Q15 (A)
Q16 (D)
Q17 (C)
Q18 (A)



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